SIMATS ENGINEERING

ASSESSMENT TOOL 1: Research Review & Analysis

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS COURSE CODE: MLA01

COURSE OUTCOME COVERED
CO4: *Develop intelligent software agents using suitable agent architectures and learning techniques.*(BTL 3)
Assessment Tool Weightage: 30%

Total Marks: 100 **Time: 90 Mins**
No. of Questions: 1

Annexure A

Research Review & Analysis

Code of Conduct:
*I (*V Sunil Kumar/ 192425292) *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action

V Sunil Kumar
Signature of the student:

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Domain Understanding	10%	
2. Literature Search and Source Selection	10%	
3. Literature Review and Synthesis	15%	
4. Comparative Analysis of AI Techniques	10%	
5. Critical Analysis of Research Findings	10%	
6. Research Gap Identification	10%	
7. Proposed Research Direction	10%	
8. Research Methodology / Analysis Framework	5%	
9. Experimental / Evidence-Based Validation	10%	
10. Result Analysis and Interpretation	5%	
11. Research Paper Quality and Referencing	10%	
12. Technical Presentation and Research Defense	5%	
Total	100%	

Signature of the Faculty Total Marks Awarded

Q1. Decision Tree Learning

Review recent research on ID3, C4.5, and CART. Compare their attribute-selection methods, pruning strategies, interpretability, computational complexity, and classification performance. Identify a research gap and suggest a possible improvement.

Title

Comparative Research Review of ID3, C4.5 and CART Decision Tree Algorithms for Classification

Abstract

Decision Tree Learning is an important supervised machine learning approach used for classification and decision-making. This review analyzes three well-known decision tree algorithms: ID3, C4.5, and CART. The algorithms are compared based on attribute-selection methods, pruning strategies, data handling capabilities, interpretability, computational complexity, and classification performance. ID3 uses Information Gain, C4.5 uses Gain Ratio, and CART mainly uses the Gini Index for classification. C4.5 improves several limitations of ID3 by supporting continuous attributes, missing values, and pruning, while CART produces binary decision trees and supports both classification and regression. Recent comparative research shows that no single algorithm is consistently best for every dataset. A major research gap is the dependence of traditional decision trees on a single splitting criterion and the difficulty of balancing accuracy, tree complexity, computational cost, and interpretability. A possible improvement is an adaptive decision tree that selects or combines splitting criteria based on dataset characteristics and applies optimized pruning.

Keywords

Decision Tree, ID3, C4.5, CART, Information Gain, Gini Index

1. Introduction

Decision Tree Learning is a supervised machine learning technique used to learn decision rules from training data. A decision tree represents decisions in a hierarchical structure containing a root node, internal decision nodes, branches, and leaf nodes. Each path from the root to a leaf can represent a classification rule.

Decision trees are widely used because their decision-making process is relatively easy to understand and interpret. They can also model non-linear relationships without requiring complex mathematical transformations. ID3, C4.5, and CART are among the most important traditional decision tree algorithms. ID3 was developed by Ross Quinlan and uses Information Gain for selecting attributes. C4.5 was developed as an extension of ID3 and introduced improvements such as Gain Ratio and support for continuous attributes. CART uses the Gini Index for classification and generally produces binary trees.

The purpose of this research review is to compare these three algorithms and identify their strengths, weaknesses, and possible areas for improvement.

2. Research Problem and Objectives

The main research problem is to determine how ID3, C4.5, and CART differ in their learning mechanisms and which characteristics make one algorithm more suitable than another for particular classification problems.

The specific objectives of this review are:

1. To study the working principles of ID3, C4.5, and CART.
2. To compare their attribute-selection methods.
3. To analyze their pruning strategies.
4. To compare their computational complexity and scalability.
5. To evaluate their interpretability and classification performance.
6. To identify limitations and research gaps in traditional decision tree algorithms.
7. To propose a possible improvement for future decision tree systems.

3. Literature Review

ID3 is one of the earliest widely used decision tree induction algorithms. It uses entropy and Information Gain to select the attribute that provides the greatest reduction in uncertainty. ID3 generally produces a multiway tree and was originally designed mainly for categorical attributes. Because it tends to grow the tree until a stopping condition is reached, pruning or other controls are important for reducing overfitting.

C4.5 was introduced as an extension of ID3. It uses Information Gain Ratio instead of simple Information Gain. This reduces the tendency of Information Gain to favor attributes having many distinct values. C4.5 can also handle continuous attributes by determining suitable thresholds. It additionally supports missing values and applies pruning to simplify the generated tree and improve generalization.

CART, or Classification and Regression Trees, uses the Gini Index for classification and generally creates binary splits. It is applicable to both classification and regression problems. Its binary-tree structure can make the tree systematic and computationally effective, although the number of levels can increase because each decision is divided into two branches.

A 2023 comparative study evaluated ID3, C4.5, CART, and other splitting approaches across multiple datasets using classification accuracy, tree depth, leaf-node count, and tree construction time. The study reported that traditional ID3 and CART performed strongly overall, with statistical testing indicating that they were preferable to several newer alternatives considered in that study.

Another comparative study examined ID3, C4.5, and CART using multiple datasets and reported differences in accuracy, execution time, and algorithm behavior. Such results indicate that algorithm selection should depend on the characteristics of the dataset rather than assuming that one decision-tree method is always superior.

Research on C4.5 also demonstrates the importance of pruning. C4.5 can apply post-pruning after constructing the tree, replacing subtrees with leaf nodes when this does not significantly reduce classification performance. This helps control tree complexity and overfitting.

Recent research has also investigated modifications to splitting criteria. For example, research on information-gain-ratio correction has examined how the gain-ratio criterion can be further improved for more balanced tree splits. This indicates that attribute selection remains an active area for improvement.

4. Comparative Analysis

The three algorithms can be compared using several important parameters.

Parameter	ID3	C4.5	CART

Parameter	ID3	C4.5	CART
Attribute Selection	Information Gain	Gain Ratio	Gini Index
Tree Structure	Multiway	Multiway	Binary
Main Use	Classification	Classification	Classification and Regression
Continuous Attributes	Limited	Supported	Supported
Missing Values	Limited	Supported	Supported
Pruning	Basic / external control	Post-pruning	Cost-complexity pruning
Interpretability	High	High	High
Computational Cost	Relatively low	Moderate	Moderate
Overfitting Control	Limited	Stronger	Strong
Splitting Strategy	Entropy-based	Gain-ratio based	Gini-based

ID3 is simple and easy to understand, but its Information Gain criterion can favor attributes with many possible values. C4.5 addresses this limitation by using Gain Ratio. CART uses the Gini Index and restricts each decision to binary splitting.

In terms of interpretability, all three algorithms provide understandable rule-based structures. However, very deep trees can become difficult to interpret regardless of the algorithm.

In terms of computational complexity, ID3 and CART can be efficient in many situations, while C4.5 performs additional processing for continuous attributes, gain-ratio calculation, and pruning. A 2023 experimental study found that ID3 and CART were significantly faster than some alternative approaches in the tested datasets.

Classification performance depends strongly on the dataset. In the 2023 study involving multiple UCI datasets, ID3 achieved an average classification accuracy of 84.74%, while C4.5 achieved 82.93% and CART achieved 84.01% across the reported experiments. These values demonstrate that the performance differences can be relatively small and dataset-dependent.

Therefore, the best algorithm cannot be selected only from the splitting criterion. Dataset size, feature type, noise, missing values, tree complexity, and computational requirements should also be considered.

5. Critical Analysis

ID3 has the advantage of simplicity. Its Information Gain criterion is easy to understand and the resulting decision rules are highly interpretable. However, its major limitation is its tendency to favor attributes with many distinct values. It also has weaker support for continuous and missing data compared with later algorithms.

C4.5 improves upon ID3 by introducing Gain Ratio, continuous-attribute handling, missing-value support, and pruning. These improvements make C4.5 more flexible for practical classification tasks. However, the additional processing can increase computational requirements, and the resulting trees may still become large or complex.

CART provides a consistent binary splitting structure and can be applied to both classification and regression. The Gini Index is computationally convenient and commonly effective for classification. However, binary splitting may require more levels than a multiway tree, potentially increasing tree depth.

A common limitation across all three methods is that tree construction is based on greedy local decisions. The best split at one node is selected without fully evaluating all possible future tree structures. Consequently, the locally optimal decision may not result in the globally optimal tree.

Another challenge is sensitivity to noisy data and small changes in the training dataset. A small change in the data can sometimes produce a different tree structure. Pruning and validation techniques are therefore important for improving generalization.

6. Research Gap

The literature shows that ID3, C4.5, and CART have different advantages, but several research gaps remain.

First, traditional algorithms generally depend on a fixed attribute-selection criterion. ID3 uses Information Gain, C4.5 uses Gain Ratio, and CART uses the Gini Index. The performance of a splitting criterion can vary according to the dataset characteristics.

Second, there is a continuing challenge in balancing classification accuracy and tree complexity. A highly complex tree may achieve strong training accuracy but can overfit the training data.

Third, the computational cost of tree construction can become important when datasets contain a large number of instances and features.

Fourth, although decision trees are considered interpretable, very large trees can reduce practical interpretability. Therefore, improving accuracy while maintaining a compact and understandable tree remains an important research problem.

Recent comparative work confirms that performance varies across datasets and evaluation measures, supporting the need for adaptive rather than one-size-fits-all decision-tree methods.

7. Proposed Research Direction

A possible improvement is to develop an Adaptive Hybrid Decision Tree that dynamically selects the most suitable splitting criterion according to the characteristics of the dataset.

The proposed approach can work as follows:

1. Analyze the dataset characteristics such as number of attributes, number of classes, feature types, missing values, and class imbalance.
2. Evaluate Information Gain, Gain Ratio, and Gini Index for candidate splits.
3. Select the criterion that provides the best balance between predictive improvement and split quality.
4. Construct the tree using the selected criterion.
5. Apply cross-validation-based pruning to control overfitting.
6. Measure accuracy, precision, recall, F1-score, tree depth, number of nodes, and training time.
7. Compare the adaptive method with standard ID3, C4.5, and CART.

This approach could improve generalization while maintaining the interpretability of the decision tree.

8. Discussion

The literature review demonstrates that ID3, C4.5, and CART remain important foundations for decision tree learning. Their main differences arise from the criteria used to select attributes and the methods used to control tree complexity.

ID3 provides a simple and interpretable approach but has limitations related to attribute bias and data handling. C4.5 addresses many of these limitations and provides greater flexibility. CART provides binary tree construction and supports both classification and regression.

Experimental evidence indicates that performance varies according to the dataset. The 2023 study comparing multiple decision-tree approaches showed that ID3 and CART performed strongly across the tested datasets, while also demonstrating that tree depth, leaf count, and construction time are useful measures in addition to accuracy.

Therefore, decision-tree selection should consider multiple factors rather than classification accuracy alone. A model with slightly lower accuracy may be preferable if it produces a much smaller and more interpretable tree with lower computational cost.

9. Conclusion

This research review analyzed ID3, C4.5, and CART decision tree algorithms with respect to attribute selection, pruning, interpretability, computational complexity, and classification performance.

ID3 uses Information Gain and provides a simple and interpretable tree but has limitations related to attribute bias and data handling. C4.5 improves upon ID3 through Gain Ratio, continuous-attribute support, missing-value handling, and pruning. CART uses the Gini Index and creates binary trees, making it suitable for both classification and regression.

The reviewed research indicates that no single algorithm consistently performs best on every dataset. The research gap identified is the dependence on fixed splitting criteria and the challenge of simultaneously optimizing accuracy, computational efficiency, tree complexity, and interpretability.

An adaptive hybrid decision-tree approach that dynamically selects the splitting criterion and uses improved pruning could therefore be a useful future research direction.

References

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Evaluation Rubrics:

Criterion	Weightage (%)	Excellent	Good	Average	Poor
1. Research Problem Identification and Understanding	10%	9–10% – Clearly identifies the research problem, objectives, scope, and relevance to AI learning approaches.	7–8% – Research problem and objectives are clearly identified with minor omissions.	5–6% – Basic understanding of the research problem with some gaps.	0–4% – Problem is unclear, inappropriate, or poorly understood.
2. Literature Search and Source Selection	10%	9–10% – Uses diverse, highly relevant, recent, and credible research papers, journals, conferences, and technical sources.	7–8% – Uses relevant and credible sources with minor limitations.	5–6% – Uses a limited number of relevant sources.	0–4% – Sources are insufficient, unreliable, or largely irrelevant.
3. Literature Review and Synthesis	15%	13–15% – Provides a comprehensive review and synthesizes findings across multiple studies rather than merely summarizing them.	10–12% – Good review with meaningful comparison of major studies.	7–9% – Basic review with limited synthesis and comparison.	0–6% – Superficial review with little understanding of existing research.
4. Analysis of AI Techniques / Methods	15%	13–15% – Provides technically accurate and in-depth analysis of relevant methods such as decision trees, statistical learning, neural networks, kernel methods, or reinforcement learning.	10–12% – Provides good technical analysis with minor gaps.	7–9% – Provides basic technical description with limited analysis.	0–6% – Methods are poorly understood or inaccurately described.
5. Comparative Analysis of Research Approaches	10%	9–10% – Critically compares approaches based on accuracy, complexity, interpretability, scalability, computational cost, and application suitability.	7–8% – Provides meaningful comparison using several parameters.	5–6% – Basic comparison with limited criteria.	0–4% – Little or no meaningful comparison.
6. Critical Evaluation of Research Findings	10%	9–10% – Critically evaluates strengths, weaknesses, assumptions, experimental methods, and findings of existing studies.	7–8% – Provides good critical evaluation with minor gaps.	5–6% – Some critical observations are provided but analysis is limited.	0–4% – Mostly descriptive with little critical evaluation.
7. Identification of Research Gaps	10%	9–10% – Clearly identifies significant research gaps, limitations, unresolved issues, and opportunities for further investigation.	7–8% – Identifies relevant research gaps with reasonable justification.	5–6% – Identifies basic gaps but provides limited justification.	0–4% – Research gaps are missing or incorrectly identified.
8. Research Insights and Recommendations	5%	5% – Provides insightful conclusions and technically feasible recommendations for future research or applications.	4% – Provides relevant conclusions and recommendations.	2–3% – Basic conclusions with limited recommendations.	0–1% – Conclusions or recommendations are absent or inappropriate.
9. Technical Report and Referencing	10%	9–10% – Report is well structured, technically accurate, professionally written, and uses a consistent citation/reference style.	7–8% – Well-structured report with minor writing or referencing issues.	5–6% – Adequate report but contains several structural or referencing issues.	0–4% – Poorly structured, inadequately referenced, or incomplete report.
10. Originality, Academic Integrity and Use of Sources	5%	5% – Demonstrates original synthesis, proper citation, and strong academic integrity with no inappropriate copying.	4% – Good originality and proper citation with minor issues.	2–3% – Some original analysis but significant dependence on sources.	0–1% – Evidence of substantial copying, poor citation, or academic-integrity concerns.